

HDOC Day-36

Menu-Driven String Operations Using Separate Functions

Problem Statement

Write a menu-driven C program to perform the following operations using separate functions:

1. Count characters in a string without using built-in length functions.
2. Print each character of a string on a new line.
3. Count vowels and consonants in a string.
4. Exit.

Algorithm

5. Start the program.
6. Read a string from the user using fgets().
7. Display a menu with four choices.
8. Read the user choice.
9. Use a switch statement to call the required function.
10. For choice 1, traverse the string until the null character and count characters, excluding the newline entered by fgets().
11. For choice 2, traverse the string and print each character on a separate line.
12. For choice 3, examine alphabetic characters. Count a, e, i, o and u (both cases) as vowels; count the remaining alphabetic characters as consonants.
13. Repeat the menu until choice 4 is selected.
14. Stop the program.

C Program

```
#include <stdio.h>

int countCharacters(char str[])
{
    int i = 0, count = 0;
    while (str[i] != '\0' && str[i] != '\n')
    {
        count++;
        i++;
    }
    return count;
}

void printCharacters(char str[])
{
    int i = 0;
    while (str[i] != '\0' && str[i] != '\n')
    {
        printf("%c\n", str[i]);
        i++;
    }
}
```

```

}

void countVowelsConsonants(char str[])
{
    int i = 0, vowels = 0, consonants = 0;
    char ch;

    while (str[i] != '\0' && str[i] != '\n')
    {
        ch = str[i];
        if ((ch >= 'A' && ch <= 'Z') || (ch >= 'a' && ch <= 'z'))
        {
            if (ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch == 'U' ||
                ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u')
                vowels++;
            else
                consonants++;
        }
        i++;
    }

    printf("Number of vowels = %d\n", vowels);
    printf("Number of consonants = %d\n", consonants);
}

int main(void)
{
    char str[100];
    int choice;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    do
    {
        printf("\n--- MENU ---\n");
        printf("1. Count characters\n");
        printf("2. Print each character on a new line\n");
        printf("3. Count vowels and consonants\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                printf("Number of characters = %d\n", countCharacters(str));
                break;
            case 2:
                printCharacters(str);
                break;
            case 3:
                countVowelsConsonants(str);
                break;
            case 4:
                printf("Exiting program...\n");
                break;
            default:
                printf("Invalid choice. Please try again.\n");
        }
    } while (choice != 4);

    return 0;
}

```

Suggested Test Input

Use the string: Hello World

Menu Choice	Expected Output	Screenshot
1	Number of characters = 11	Add terminal screenshot here
2	H, e, l, l, o, space, W, o, r, l, d printed on separate lines	Add terminal screenshot here
3	Number of vowels = 3; Number of consonants = 7	Add terminal screenshot here
4	Exiting program...	Can be included with the last screenshot

Result

The menu-driven C program was implemented successfully using separate functions for counting characters, printing characters individually, and counting vowels and consonants in a string.

Actual Output Screenshots

```
(kali㉿kali)-[~/Desktop/apple]
$ ./hdoc36
Enter a string: Hello World

— MENU —
1. Count characters
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 1
Number of characters = 11
```

```
— MENU —
1. Count characters
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 2
H
e
l
l
o

W
o
r
l
d
```

— MENU —

1. Count characters
2. Print each character on a new line
3. Count vowels and consonants
4. Exit

Enter your choice: 3

Number of vowels = 3

Number of consonants = 7

— MENU —

1. Count characters
2. Print each character on a new line
3. Count vowels and consonants
4. Exit

Enter your choice: 4

Exiting program...